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Applied Robotics (FRTN85) Applied Robotics for Architects (FRTN80)

SUGGESTED PROJECTS HT 2026



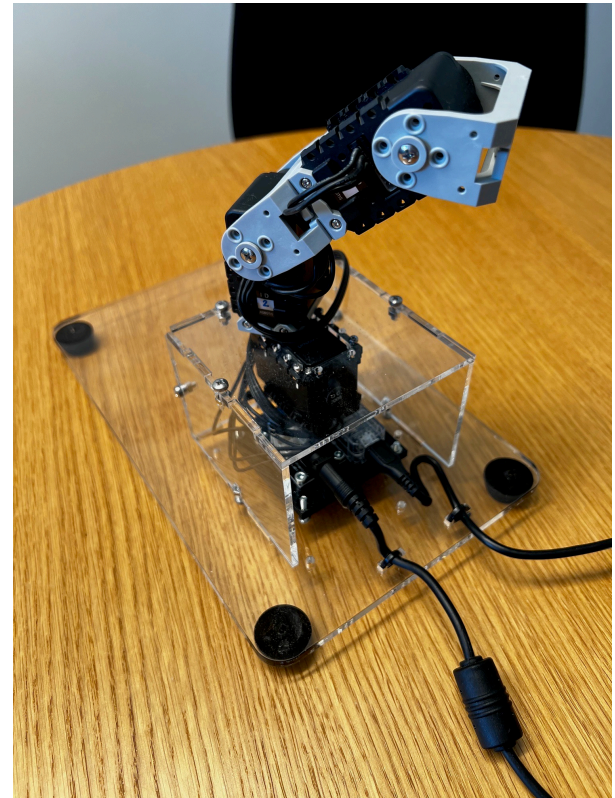
Sign-up for a Course Project

HT2026:

- The following presentation contains a set of briefly described project proposals for FRTN85 & FRTN80 (denoted A to F)
- The projects will be made in **groups of approx. 6 students**, and an advisor will be assigned to each group
- Please submit a list of your **top three alternatives in Canvas before midnight on Thursday September 17, 2026**
- You can also add names of group members in case you already know other course participants to work with on the same topics
- Only one Canvas submission per group is necessary

A: 3-DoF Dynamixel Robot

- Three rotational Dynamixel servos are the joints of a **3-DoF robot arm**
- Can easily be controlled by sending joint angle references through Python
- Can be **borrowed and brought home**
- Examples: Pour rice with a spoon, drawing, bead sorting, learning control, etc.



B: Drawing or Sand-Shaping Robot

- **Alt. I: Drawing project** on the ABB IRB140 robot: Motion generation from photo / picture and calibration with sensors on tilted drawing plane
- **Alt. II: Sand-shaping project** on the ABB IRB140 robot: design, develop tools, and potentially include sensors for feedback



[<https://www.instructables.com/id/Drawing-Robot/>]



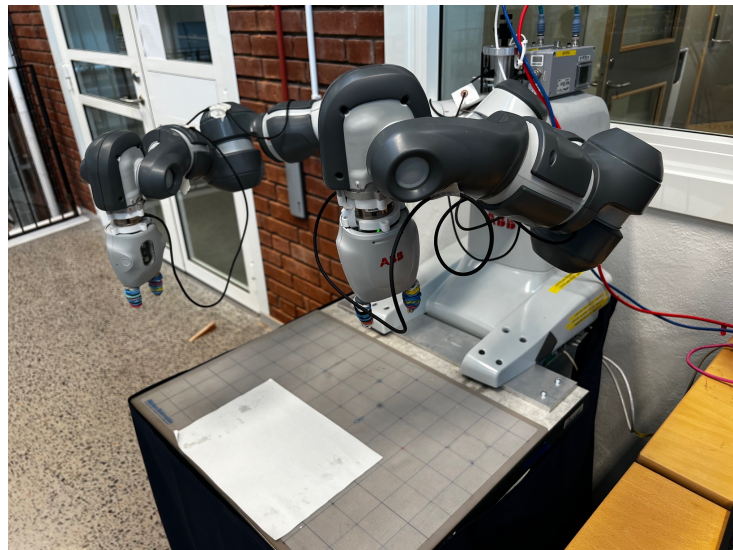
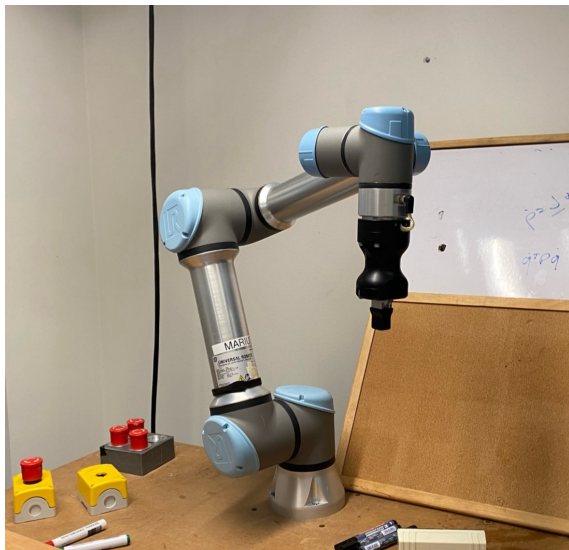
[Gramazio Kohler Research, ETH Zürich
<https://gramaziokohler.arch.ethz.ch/web/lehre/e/0/0/0/208.html>]



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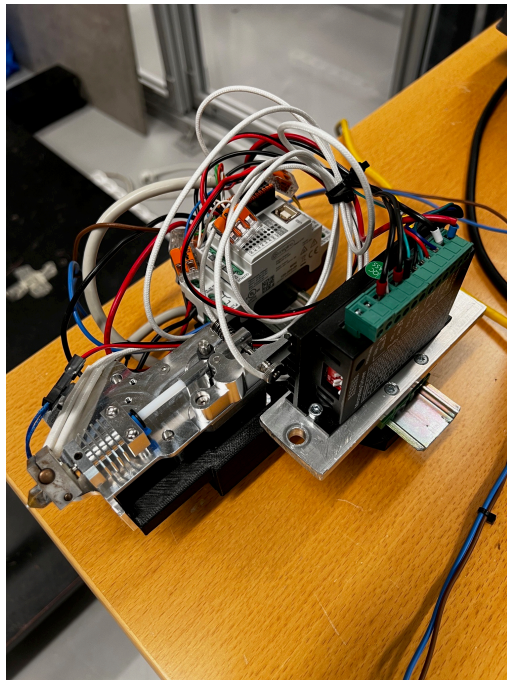
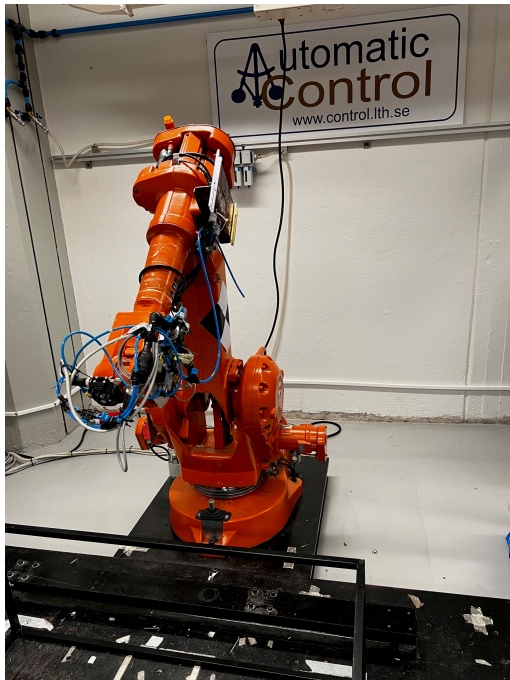
C: UR5e or YuMi Robot Control with Workspace Sensing

- UR5e or YuMi robot equipped with **force sensing** for interaction tasks or **camera for vision**
- Python interface to programming of UR5e, Python or RAPID for YuMi
- Examples: Solve Rubick's cube, play tic tac toe or chess, move rice with spoon, etc.



D: 6D-printing with IRB2400

- Fabrication with robots using **6D-printing** with ABB IRB2400 (including calibration of tool / workpiece).
- Contains many course components from FRTN80.



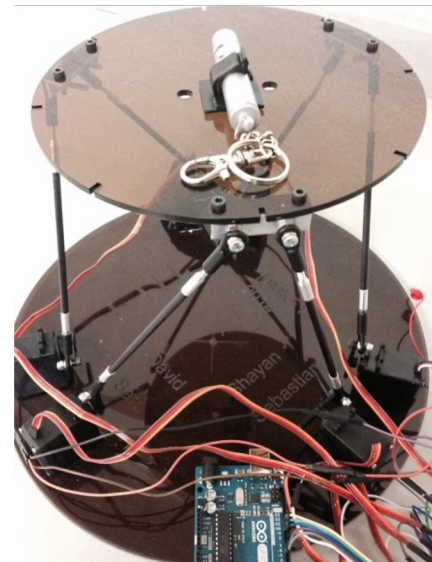
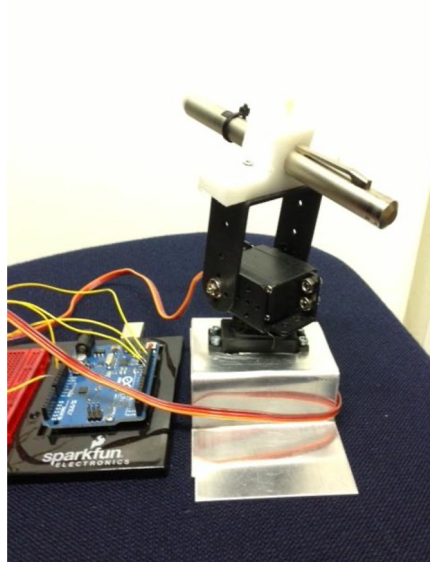
Example robot 3D-printing from ETH, Switzerland



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E: Arduino-Based Balancing Robot

- **Self-balancing robot** via an Arduino (or Raspberry)
- Develop kinematics and control, e.g., using IMU sensor measurements
- Previous experience with Arduino recommended



<https://www.youtube.com/watch?v=QdKo9PYwGaU>



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F: Your Own Suggestion

- You are also welcome to **suggest a robot project of your own choice**
- For example, a robot you have at home, machine learning for robots, iterative learning control, building your own robot, etc.
- We will then evaluate if it is possible to realize it within the course, and if so, assign a project advisor

